

Exhaustive Guide to Managing Grapevine Esca (Black Measles)

1. Overview

Grapevines are highly resilient plants, often living and producing fruit for many decades. A vineyard is a long-term investment, designed to be passed down through generations. However, maintaining a healthy and productive vineyard requires constant vigilance against a hidden, internal threat known collectively as Grapevine Trunk Diseases. Among the most destructive, complex, and mysterious of these trunk diseases is Esca, which is commonly referred to in the United States by the highly descriptive name "Black Measles".

Unlike common plant ailments that exist on the surface of the leaves and can be easily washed away, Esca is a disease of the deep, internal wood. It is not caused by a single type of bug, nor is it caused by a single, simple germ. Instead, Esca is a complex syndrome caused by a succession of different fungal species that invade the plant over many years. These microscopic fungi slowly decay the internal heartwood of the vine, ultimately destroying the plant's ability to transport water and essential nutrients from the roots up to the leaves and fruit.

The history of Esca is as fascinating and complex as the disease itself. More than a century ago, in 1898, researchers and farmers in France and California first began documenting a mysterious disorder that was devastating mature grapevines. At the time, agricultural science had not yet discovered that fungi could rot a living plant from the inside out. In France, they called this terrifying phenomenon "folletage," and in California, it was simply known as "sunstroke". This was because seemingly healthy vines would suddenly collapse, wither, and die during the absolute hottest days of the summer season. Farmers logically, but incorrectly, believed that the intense heat of the sun had simply baked the plants to death.

It took decades of painstaking research to uncover the truth. In 1909 and 1912, scientists looking through microscopes finally observed fungal threads, known as mycelium, deep inside the tissues of the diseased and dying vines. They realized that the true cause of the "sunstroke" was a hidden, internal decay that was eating away at the inside of the trunk, blocking the water supply exactly when the plant needed it most.

Today, Esca is universally recognized as a massive global threat to viticulture, capable of causing devastating yield losses of up to 50 percent in individual vines. It is a particularly frustrating and heartbreaking disease for growers to manage because a grapevine might be infected for many years without showing a single outward sign of illness. The fungi can live quietly inside the hard wood, acting as harmless, silent residents. They remain dormant until the vine experiences an episode of severe environmental stress—such as a prolonged drought, poor, waterlogged soil conditions, or the burden of producing an overly heavy crop of fruit. Once triggered by this stress, the disease violently attacks the plant's internal water pipes, leading to severe leaf discoloration, ruined and unmarketable fruit, and sometimes the sudden, shocking death of the entire plant.

While Esca is generally considered a disease of older, mature vines—typically making its first visible appearance when a vineyard is five to seven years old—the initial fungal infections usually begin much, much earlier. These silent infections often occur when the vines are just baby plants being grafted in a nursery, or when they are first planted and pruned in their early

establishment years in the field. A related, aggressive form of the disease, known as Petri disease or "young vine decline," affects newly planted vineyards, drastically stunting their growth right from the start and preventing them from ever reaching maturity. Because the Esca fungi live deep inside the hard, structural wood of the trunk, the disease cannot be cured with a simple chemical spray applied to the leaves. Once the internal wood is rotted, the decay is permanent. Therefore, managing Esca requires a comprehensive, year-round, holistic strategy. It requires a deep understanding of plant health, focusing heavily on prevention, extremely careful pruning techniques, immediate wound protection, and reducing overall plant stress to keep the vines as strong and healthy as possible. This exhaustive guide will provide a detailed, easy-to-understand breakdown of how to identify, treat, and prevent Esca to protect the longevity, health, and productivity of your vineyard.

2. Symptoms

Identifying Esca in the field can be an incredibly tricky and frustrating process because the visual symptoms are highly inconsistent and unpredictable. A vine might look terrible and sickly one year, appear completely healthy and robust the next year, and then suddenly collapse and die the following year. This baffling, unpredictable nature is believed to be closely tied to the weather and how subtle environmental changes trigger the fungi hidden inside the wood to release harmful toxins into the plant's sap stream. However, when symptoms do eventually appear, they are usually very distinct, highly visual, and can be categorized into four main areas: the leaves, the fruit, the internal wood, and the phenomenon of sudden collapse.

Leaf Symptoms (The "Tiger Stripe" Pattern)

The most famous, prominent, and easily recognized sign of Esca occurs on the foliage of the vine. Usually appearing mid-to-late in the growing season, particularly during the hot months of July and August, the mature leaves begin to show distinct, small, discolored patches. These patches always appear between the major veins of the leaf, rather than on the veins themselves.

The color of these sickly patches depends entirely on the type of grape you are growing. In white grape varieties, these interveinal patches turn a pale, sickly yellow. In red grape varieties, they turn a deep, dark red or a vibrant dark pink. As the disease progresses inside the plant, these brightly colored patches begin to dry out, die, and turn into crispy, brown, dead tissue, a process known as necrosis.

Because the tissue directly next to the leaf veins remains green and alive for a longer period, the leaf takes on a striking, highly visible striped appearance. Farmers and horticulturists commonly refer to this as the "tiger stripe" pattern. Eventually, as the damage becomes too severe, the entire leaf may dry up completely, curl at the edges, and drop off the vine prematurely, leaving the fruit exposed to the sun. Interestingly, these dramatic leaf symptoms might only appear on a single shoot or a single branch of the vine, while the rest of the plant looks perfectly normal and healthy.

Fruit Symptoms (Black Measles)

Esca also severely impacts the grapes themselves, which is what gives the disease its common North American name: Black Measles. At any point between the time the fruit first sets on the

vine and the time it begins to ripen, the individual berries may develop small, round, dark, highly visible spots.

These spots are typically dark purple, deep brown, or a purplish-gray color, and they are often surrounded by a darker, defined ring, making them look exactly like a measles rash on human skin. This spotting is much easier to see on green or white grape varieties, where the dark spots contrast sharply with the light skin of the fruit.

In severe cases of Esca, the damage goes far beyond cosmetic spotting. The affected berries will physically crack open, shrivel up, dry out, or become highly susceptible to secondary rotting and spoilage. Vineyards suffering from a severe outbreak of "measles" will experience heavy crop and financial losses, as the fruit is visually unappealing, fails to mature or sweeten properly, and is rejected by winemakers. It is important to note that symptomatic, spotted fruit is usually only found on shoots that also have the tiger-striped leaves, but you can sometimes find shoots with diseased leaves and perfectly clean fruit.

Internal Wood Symptoms (The Hidden Decay)

To truly and definitively confirm the presence of Esca, you have to look deep inside the plant. If you take a saw or a sharp pair of pruning shears and cut a cross-section through a diseased branch, spur, or the main trunk, you will see clear, undeniable evidence of the fungal invasion. A healthy grapevine trunk should be uniformly light-colored and clear inside. An Esca-infected trunk, however, will show dark brown or stark black spots arranged in distinct concentric rings. These look like dark speckles or heavy streaks running vertically through the wood.

This dark streaking is actually the physical manifestation of the plant's internal water-carrying vessels (called the xylem) becoming plugged up. When the vine detects the invading fungus, it panics and tries to block the infection from spreading by filling its own water pipes with a dark, amber-colored, sticky, gummy substance, a process called gummosis. The plant also produces balloon-like growths called tyloses that swell up inside the pipes to block them. While this defense mechanism tries to trap the fungus, it tragically clogs the plant's plumbing, leading to drought stress.

In older, mature vines suffering from advanced, late-stage Esca, a completely different type of fungus takes over the wood. This secondary fungus turns the hard, strong central heartwood of the trunk into a soft, spongy, crumbly, whitish-yellow mass of rot. When this soft, spongy decay is present, the vine is suffering from what plant pathologists call "esca proper".

Stunted Growth in Young Vines (Petri Disease)

When the Esca fungal complex attacks very young vines—often those aged between one and five years old—it manifests differently and is known as young vine decline or Petri disease.

These young vines simply fail to thrive after being planted.

They show severely stunted growth, tiny trunk diameters that fail to thicken, abnormally short spaces between the leaves on the shoot (a condition known as shortened internodes), and very sparse, unusually small foliage. Because their internal plumbing is compromised from the very beginning, these young vines may slowly waste away over a period of a few years, never becoming strong enough to produce a viable crop.

Sudden Collapse (Apoplexy)

The most terrifying and dramatic symptom of the Esca complex is a phenomenon known as

apoplexy. In this tragic scenario, a mature, producing vine that appeared completely healthy the day before, or perhaps only showed very minor leaf spotting, will suddenly and completely collapse.

Over the course of just a few days, the leaves will rapidly wilt, turn crispy, and die, while the fruit will instantly shrivel into raisins. This violent event almost always happens in the absolute middle of summer, particularly when a period of heavy rainfall is immediately followed by a blast of extremely hot, dry weather. The sudden, intense heat causes the vine's leaves to demand massive amounts of water to stay cool. But because the internal pipes are severely clogged by the disease and the defensive gums, the plant cannot pull enough water from the roots. The vine effectively suffocates from a lack of moisture and suffers a sudden, fatal, irreversible hydraulic "stroke".

3. Causes & Spread

Understanding how Esca works requires understanding that it is a "complex" disease. This means it is not caused by a single bug, nor is it caused by a single germ or a single fungal spore. Instead, it is caused by an entire diverse community of different fungal species that attack the vine in a highly specific sequence, often taking many years to complete their destructive work.

The Fungal Invaders and Succession

The infection process usually begins with a group of fungi known as the "pioneer" colonizers. The most common and widespread of these initial invaders is a fungus called *Phaeomoniella chlamydospora*, closely followed by dozens of species of another fungus called *Phaeoacremonium* (with *Phaeoacremonium minimum* being the absolute most prevalent worldwide). These pioneer fungi are the first to enter the wood. They begin the slow process of irritating the vascular system, causing the plant to plug its own pipes with dark gums, which leads to the dark brown streaking seen in the wood.

Years later, after the pioneer fungi have weakened the wood and altered its internal chemistry, a second wave of entirely different fungi arrives to finish the job. These are called basidiomycete fungi, and they belong to the exact same broad family of fungi that produce common forest mushrooms. In fact, if you look very closely at the trunk of a severely decayed vine in an old vineyard, you might even see small, hard, shelf-like mushrooms growing directly out of the bark. The most famous of these secondary invaders is *Fomitiporia mediterranea*. This second group of fungi is solely responsible for the aggressive, deep rotting that turns the hard, structural wood of the grapevine into a soft, spongy, white mess, completely destroying the integrity of the trunk.

How the Disease Spreads in the Vineyard

The fungi that cause Esca and other grapevine trunk diseases live deep inside the dead and diseased wood of infected grapevines all year round, quietly surviving the changing seasons. The primary way the disease spreads from a sick, dying vine to a healthy, productive vine is through the air, specifically during the cold, wet winter pruning season.

Hidden inside the dead wood of the trunk, the fungi form microscopic, overwintering survival structures (called perithecia or pycnidia) that act like tiny, highly efficient spore factories. When the weather turns cold and wet—specifically during winter rain, snow, or even heavy, lingering

winter fog—these factories burst open and release millions of microscopic fungal spores into the surrounding air. The winter wind catches these spores and blows them across the entire vineyard.

At the exact same time of year, vineyard workers are out in the field, working hard to heavily prune the dormant vines to prepare them for the upcoming spring growth. Every single time a worker makes a cut with their sharp pruning shears, they are amputating a piece of the plant and creating a fresh, open, bleeding wound. If a wind-blown Esca spore lands on that freshly cut wood, the fungus is immediately sucked inside the plant's vascular system.

Because the vine is deep in winter dormancy, its immune system is essentially asleep, and the sap is not flowing. This means the wound takes a very, very long time to heal. In the freezing middle of December or January, a simple pruning cut might remain completely open and highly vulnerable to fungal infection for two to three full weeks. In some severe cases, depending on the weather, wounds can remain susceptible to airborne spores for up to four months. Once the fungal spore gets inside the open wound and establishes itself deep in the wood, it becomes a permanent resident. It cannot be washed off, and it cannot be killed with traditional foliar sprays applied to the leaves.

Spread Through the Nursery and Planting Material

Wind and winter rain are not the only culprits. Because these complex fungi can live inside the vine for many years without showing any outward tiger-stripe symptoms, the disease is frequently and unknowingly spread by human hands through the propagation and sale of new baby plants.

When agricultural nurseries create new grapevines to sell to farmers, they take live wood cuttings from older "mother" vines. If the mother vine is secretly infected with Esca fungi, the cuttings taken from it will also be deeply infected from day one. Furthermore, the microscopic fungi can be rapidly spread from one cutting to a thousand cuttings on the sharp blades of the grafting machines used to fuse the top of the vine to the rootstock. The disease can also spread rapidly in the large hydration tanks where thousands of cuttings are soaked together in water before planting.

If young, nursery-grown vines undergo severe environmental stress during cold storage or transport in shipping trucks, their weak, immature immune systems collapse. This allows the small amount of fungus hiding inside to multiply rapidly. When the hopeful farmer eventually plants these stressed, infected babies in the field, they will inevitably develop Petri disease, show stunted growth, and fail to ever produce a profitable crop.

The Massive Role of Environmental Stress

Perhaps the most fascinating, and frustrating, aspect of the Esca complex is that the pioneer fungi are broadly considered to be "endophytic". In biology, an endophyte is a microscopic organism that can live inside a plant for its entire life without causing any disease or harm whatsoever. In major wine regions like California and Oregon, exhaustive scientific surveys have shown that these specific fungi are present in almost every single vineyard aged between 10 and 80 years old. Yet, not every single vineyard dies from Esca.

The disease only becomes a rapid killer when the vine experiences extreme, compounding stress. The most significant and universally recognized trigger is water stress. When a vine is suffering from a summer drought, it desperately struggles to pull water from the dry soil. If the internal pipes of the trunk are already partially blocked by the pioneer fungi and the plant's own

defensive gums, the minor water shortage becomes an absolute crisis. The vine panics, and the lack of water and nutrients causes the severe, brightly colored "tiger stripe" symptoms to burst forth on the leaves as toxins are pulled up from the rotting wood.

Other severe environmental stressors that can trigger the disease include poor soil drainage (which drowns the roots), heavily compacted soil (which suffocates the roots by denying them oxygen), planting the baby vines in poorly prepared, shallow holes (causing the roots to bend backward in a suffocating "J-shape"), general soil malnutrition, and allowing very young, immature vines to carry a massive, heavy crop of fruit before their root systems are fully established. Any factor that weakens the plant gives the sleeping fungi the upper hand.

4. Treatment / Remedies

Because the Esca disease complex is located deep inside the hard, woody, structural trunk of the vine, there is absolutely no magic chemical that can be sprayed onto the green leaves to cure it. Once the internal wood is rotted and turns to that white, spongy mess, it stays rotted forever. However, brilliant farmers, horticulturists, and scientists have developed a variety of highly effective, proven strategies to protect fresh pruning wounds, stop new infections from occurring in the winter, and manage the health of vines that are already sick. These treatments are broken down into natural, organic home remedies and synthetic chemical remedies.

Natural / Home Remedies

For organic farmers, passionate home gardeners, or commercial growers looking to reduce their reliance on harsh chemicals, there are several powerful, natural strategies to protect vines and manage the disease.

- **Homemade Pruning Pastes:** Long before modern synthetic chemicals were invented, traditional farmers protected their valuable fruit trees and grapevines using thick, nutrient-rich, homemade pastes. These pastes act as hard physical barriers against the wind-blown winter spores. They seal the fresh pruning wound, keep the freezing water out, and introduce beneficial, healthy microbes to the cut surface to outcompete the bad Esca fungi.
 - *The Biodynamic Tree Paste:* This old, highly respected farming tradition uses incredibly simple, natural materials to heal cuts, renew the tree trunk, and give the plant a burst of healthy energy. To make this powerful paste, you mix equal parts of fresh, organic cow manure, potter's clay (or bentonite clay), and silica sand (or diatomaceous earth). You slowly add rainwater or spring water and stir the mixture deeply until it forms a thick, runny, paintable paste. The clay in the mixture acts as a hard, impenetrable physical barrier as it dries in the sun. The fresh, organic cow manure is packed with billions of healthy, beneficial microbes that quickly colonize the open sap of the wound, leaving absolutely no physical room or food for the bad Esca spores to land and grow. Finally, the sand and silica help dry out the wound quickly, speeding up the plant's natural healing process. To apply it, simply use a stiff brush to scrub the bark clean of moss and overwintering insects, and then generously paint the thick slurry directly onto the fresh pruning cuts with a standard paintbrush.
 - *The Horsetail Tea Variation:* Some dedicated organic growers enhance the standard clay and manure paste by brewing a strong tea. They do this by boiling

the dried horsetail plant in clean rainwater for a full hour, letting it cool and steep overnight, and then using that nutrient-dense tea instead of plain water to mix the clay and manure. The horsetail plant is naturally very high in silica, which biologically strengthens the paste and helps it form a tight, long-lasting barrier on the vine for an entire year.

- *The Beeswax and Linseed Oil Sealant*: For a cleaner, wax-based approach that doesn't involve manure, you can easily make a paste that physically repels winter rain and kills spores on contact. Set a pot of water to boil on the stove and place a heat-safe metal or glass bowl over it (creating a double boiler). Pour about 180 ml of organic olive oil or raw linseed oil into the bowl. Once the oil is warm, slowly grate in about 45 grams of raw, natural beeswax. As the wax melts completely into the oil, remove the bowl from the heat and gently stir in 15 to 20 drops of natural, highly concentrated essential oils—such as tea tree oil, lavender oil, mānuka oil, or rosemary oil. These specific essential oils contain powerful, natural antiseptic and antibacterial properties that literally destroy fungal spores on contact. Let the warm mixture cool in a glass jar until it sets into a soft, spreadable wax. When smeared over a fresh pruning cut, the beeswax creates a permanent, waterproof layer that absolutely prevents any pathogens from entering the plant's vascular system.
- **Commercial Natural Sealants (VitiSeal and InocBloc)**: If you are running a large commercial operation and do not have the time to mix your own paste in buckets, there are highly effective, pre-mixed organic products available on the market.
 - *VitiSeal*: VitiSeal is a brilliant natural, water-based product made from a proprietary blend of essential oils and plant-based polymers. It is completely free of toxic chemicals, volatile organic compounds, and has no foul chemical odor. It is so thin and easy to use that it can be diluted with water (mixing 1 part VitiSeal concentrate to 9 parts water) and sprayed directly out of a backpack sprayer or a tractor-mounted sprayer onto the pruned vines. Once sprayed, it dries rapidly in just one to two hours, forming a tight, elastic, rubbery barrier over the wound that physically stops fungi from entering the wood.
 - *InocBloc*: Another highly effective, commercially available natural product is InocBloc, which is a thick, organic pruning paste made entirely from natural pine tar. Pine tar has been used by sailors and farmers for centuries to naturally seal wood against rot. It has proven highly effective at preventing severe trunk diseases in grapevines, kiwis, and apple orchards around the world.
- **Biocontrol Fungi (The Trichoderma Army)**: One of the most fascinating and scientifically advanced natural remedies involves fighting bad fungi with good fungi. There are incredibly beneficial, microscopic soil fungi known generally as *Trichoderma* (specifically utilizing strains like *Trichoderma asperellum*, *Trichoderma gamsii*, and *Trichoderma atroviride*). These good fungi are incredibly aggressive natural predators. When you spray a solution of Trichoderma onto a fresh, bleeding pruning wound, they rapidly colonize the cut wood, consuming the plant's sap. When the winter winds eventually blow the bad Esca spores onto the wound, the Trichoderma fungi literally attack and eat them, protecting the plant naturally through biological warfare. Commercial products containing these living Trichoderma spores (such as Bio-Tam 2.0 or Vintec) must be applied very shortly after pruning (ideally within 24 hours) to give the good fungi enough time to establish their colony before the bad fungi arrive on the wind.
- **Trunk Renewal (Vineyard Surgery)**: If a grapevine is already deeply infected with Esca and is showing severe tiger-stripe leaf symptoms or fruit rot, you do not necessarily have

to rip the entire plant out of the ground with a tractor. Because the Esca infection almost always comes from the top down—entering through the winter pruning cuts on the upper branches—the root system hiding deep underground is often perfectly healthy and incredibly strong. To save the plant, you can perform a radical agricultural procedure known as "trunk renewal". This involves taking a heavy saw and cutting the main, thick trunk of the vine completely off, very low to the ground, well below any internal visual signs of the dark, diseased, streaked wood. It is absolutely crucial to inspect the stump and ensure you have cut down into clean, pure white, healthy wood. Do not attempt mere "cosmetic surgery" where you just try to scrape the dark spots out of the wood; the entire infected section must be forcefully removed from the vineyard. Once the sick upper trunk is gone, the massive, healthy root system will push up a brand new, incredibly vigorous green shoot (called a sucker) from the base of the stump. You can easily tie this fast-growing new shoot to the trellis wire and retrain it over the next year to become the new trunk. Within just one to two years, the vine will be back to full, robust fruit production, entirely free of the devastating disease.

- **Biostimulants (The AF5 Breakthrough):** Recent, highly exciting scientific advancements have introduced a completely new class of treatments called biostimulants. A specific, highly researched foliar spray formulation known as AF5 (which contains a unique blend of potassium acetate, alpha-tocopherol, and propylenic glycol) has shown incredible, consistent promise in field trials. Unlike traditional fungicides that attempt to poison and kill the pathogen directly, AF5 is sprayed gently onto the green leaves during the summer growing season to deeply boost the plant's own internal immune system. Exhaustive RNA sequencing studies show that spraying AF5 on sick, symptomatic vines directly activates the plant's defense genes (specifically receptor-like kinases, WRKY transcription factors, and the ethylene and jasmonate signaling cascades). Furthermore, it physically improves the sick plant's ability to open its leaf pores (stomatal conductance), allowing it to sweat and cool itself (transpiration) much more effectively, which alleviates the drought stress caused by the clogged trunk. Across multiple years of study, AF5 has significantly reduced the severity of the tiger-stripe symptoms by up to 50 percent. It stands as one of the only treatments in the world that actually helps a vine *after* it has already started showing severe leaf symptoms.

Chemical Remedies

For massive, thousand-acre commercial vineyards where hand-painting beeswax pastes on millions of individual cuts is simply too labor-intensive and financially impossible, synthetic chemical fungicides remain the absolute standard method for protecting pruning wounds from the airborne spores that cause Esca.

- **Fungicide Sprays (Topsin M and Rally):** The most universally recommended, scientifically backed chemical protection protocol involves spraying all fresh pruning cuts with a potent combination of two powerful, specialized fungicides: thiophanate-methyl (commonly sold under the brand name Topsin M) and myclobutanil (commonly sold under the brand name Rally). Topsin M is a highly effective systemic fungicide. This means that when it is sprayed onto the cut wood, it doesn't just sit on the surface; it actually gets absorbed deep into the wood and moves through the plant tissues to provide deep, lasting protection against wood-decaying fungi. When physically combined in the spray tank with Rally, it creates a formidable, broad-spectrum chemical shield against a massive variety of Esca spores.

- **Application Timing is Everything:** The absolute secret to success with chemical fungicides is speed. Pruning wounds are at their most vulnerable the exact moment they are cut by the shears. Therefore, liquid fungicides must be applied as soon as humanly possible, with agricultural extensions highly recommending application within 24 hours of making the pruning cut. If a grower waits longer than a single day, the winter wind may have already blown the microscopic spores deep into the wood's pores where the chemical spray simply cannot reach them to kill them. The most efficient method is to have one crew member rapidly cutting the vines, while a second person follows closely behind with a targeted backpack sprayer or a specially designed directed tractor sprayer, thoroughly soaking the cut ends of the wood before they can dry out.
- **Dormant Surface Sterilants (Lime Sulfur):** Another highly aggressive chemical approach is to spray the entire dormant vineyard with a heavy, soaking dose of calcium polysulfide, which is commonly known as Lime Sulfur or sold under the brand name Sulforix. This is done immediately after the grueling winter pruning is entirely finished. Lime sulfur is a harsh chemical that acts as a total surface sterilizer. It physically burns and kills any dormant fungal spores resting on the rough bark or lingering on the fresh cuts, effectively sanitizing the entire vineyard block before the warm spring rains arrive.

5. Preventive Measures

Because curing a fully rotted, mature trunk is biologically impossible, the absolute best, most cost-effective way to manage Esca is to prevent the complex fungi from ever entering the wood in the first place. This requires a paradigm shift, altering how and exactly when the vineyard is managed, starting from the very day the baby vines are planted in the dirt.

- **Delayed Pruning:** As previously discussed, winter pruning creates wide-open, bleeding wounds, and winter rain easily spreads the fungal spores. Furthermore, when it is freezing cold in the dark days of December and January, the vine's internal metabolism is completely asleep, meaning a simple pruning wound might take three full weeks to heal and scab over naturally. A simple, entirely free, but highly effective preventive measure is a practice called "delayed pruning." Instead of rushing to prune in the dead of winter, growers simply wait until late February or March when the weather is noticeably warmer and the heavy winter rains are beginning to stop. In these warmer spring temperatures, the vine's sap is actively flowing, and a fresh pruning wound can completely heal and seal itself in just one or two short days, drastically reducing the window of time that the plant is vulnerable to airborne infection.
- **Double Pruning (Pre-Pruning):** In massive commercial vineyards, it is simply impossible to wait and prune thousands of acres in a single week in March. The ingenious solution to this logistical nightmare is a technique called "double pruning." In the middle of winter, a crew or a large tractor-mounted machine goes through the vineyard and roughly chops off the bulk of the long, tangled summer growth. However, instead of cutting all the way down to the permanent wood, they purposefully leave a long, temporary stub of wood (about 12 to 15 inches long). If fungal spores land on these high winter cuts and cause an infection, the infection will slowly try to travel down that 15-inch stub. Then, in late spring when the weather is beautifully warm and dry, workers go back through the vineyard with hand shears for a rapid "finishing pass." They snip off those 15-inch stubs, completely cutting away the winter infections and leaving a fresh, clean, permanent cut right next to the trunk. Because the weather is now warm, this final, important cut heals instantly, trapping

the disease in the discarded wood.

- **Sanitation and Tool Hygiene:** The simplest way to stop spreading a disease is to keep things meticulously clean. Whenever a vineyard worker is pruning a vine that looks sickly, dead, or has dark, suspicious spots inside the wood, they must immediately stop and disinfect their pruning shears with alcohol or a bleach solution before moving to the next healthy plant. Furthermore, all the dead wood and chopped-off infected branches must be physically dragged out of the vineyard and burned or destroyed. Leaving dead, infected wood laying on the damp ground between the vineyard rows just provides a perfect breeding ground for the fungi to release millions more spores next winter.
- **Nursery Prevention (Hot Water Treatment):** Prevention must start before the vines are even planted in the soil. Agricultural nurseries must take extreme care not to accidentally sell infected baby plants to hopeful farmers. When making young cuttings, nurseries should dip the cut wood in protective waxes that contain powerful fungicides or plant growth regulators to seal the ends and promote rapid healing. More importantly, dormant baby cuttings can be fully submerged in a massive hot water bath (heated to precisely 122°F or 50°C) for exactly 30 minutes. This highly specific hot water treatment is warm enough to cook and kill the fungal spores hiding deep inside the bark, but cool enough that it does not damage or kill the sleeping baby plant.
- **Choosing Resistant Varieties:** Not all grapevines are equally susceptible to the Esca complex. The deep genetics of the plant play a massive, undeniable role in how easily the fungi can destroy the wood. Science has shown that some grape varieties have naturally thicker, stronger wood (featuring a much higher lignin content) and much narrower internal water pipes (smaller xylem vessel diameters). These tiny, incredibly tough pipes make it very difficult for the fungi to spread vertically through the plant. For example, popular varieties like Merlot and Arneis have thick lignin and narrow pipes, making them highly resistant and much less likely to ever show Esca symptoms. On the other hand, varieties like Cabernet Sauvignon and Glera have very wide internal pipes and lower lignin, making them incredibly vulnerable to rapid fungal infection and severe, fast decline. If a farmer is planting a new vineyard in a wet, humid area known for trunk diseases, choosing a tougher, genetically less susceptible variety is a critical first step to ensuring the vineyard survives for decades.

6. Suggestions & Best Practices

Beyond direct chemical treatments and clever pruning tricks, the overall lifestyle, nutrition, and daily management of the vineyard play a massive role in keeping Esca at bay. Remember, these complex fungi often live peacefully inside the plant until the vine gets severely stressed.

Therefore, keeping the vine happy, perfectly hydrated, and well-fed is your ultimate, ongoing defense mechanism.

- **Mastering Post-Harvest Irrigation:** One of the most common and devastating mistakes farmers make is completely abandoning their vines after the grape harvest is finished in the fall. Once the heavy fruit is picked, the vine is physically exhausted. It desperately needs deep soil water to keep its green leaves functioning so it can perform photosynthesis and store up essential carbohydrates in its roots for the long, cold winter. If the soil is bone dry after harvest, the vine cannot pull in nutrients, leading to dry, weak, brittle woody tissues. A vine that goes into winter weak and dehydrated will suffer massive physiological stress when it wakes up in the spring. This stress is the exact trigger that

causes the sleeping Esca fungi to go on the attack. Always provide a deep, thorough watering immediately after the harvest is complete to ensure the vine builds up strong, robust winter reserves.

- **Careful, Balanced Fertilization (The Nitrogen Paradox):** Feeding your vines is critical for a good crop, but when it comes to managing Esca, more fertilizer is absolutely not always better. In fact, deep agricultural research has shown a surprising, dangerous twist: feeding an infected vine excessive amounts of nitrogen fertilizer or heavy trace microelements can actually cause the vine to break out in severe "tiger stripe" leaf symptoms. Why does this happen? Because heavy nitrogen applications cause the vine to rapidly grow a massive, lush canopy of giant green leaves. A giant canopy of leaves naturally demands a massive amount of water to stay cool. If the internal trunk pipes are already partially clogged by the Esca fungus, the giant leaves will try to suck up water that simply cannot make it through the blocked trunk. The result is an instantaneous, severe drought stress within the plant, causing the leaves to burn, scorch, and die almost immediately.
- **Standard Fertilizer Guidelines:** Instead of over-feeding to try and push the vine to grow faster, aim for perfect, gentle balance. Always conduct a proper soil test first to see what the dirt actually needs. If no soil test is available, standard, highly sustainable practices for grapevines suggest applying a gentle, balanced 10-10-10 fertilizer (which contains equal parts Nitrogen, Phosphate, and Potash). For newly planted baby vines, apply just 2 ounces (a quarter cup) per vine when spring growth begins, spread evenly in a 4-foot circle around the base. By the time the vine is mature (three years or older), apply 1/2 pound of 10-10-10 per vine exactly at bud-break. Keeping the nutrition steady and completely avoiding explosive, nitrogen-fueled canopy growth will keep the vine's water demands highly manageable and keep the Esca symptoms hidden away.
- **Mind the Calcium:** Advanced scientific studies measuring the chemistry of sick leaves have noted that when Esca symptoms explode in the summer, the calcium levels inside those damaged leaves absolutely plummet. Calcium is an essential building block for creating strong plant cell walls and fighting off severe diseases. Ensuring your soil has adequate calcium levels, and that the soil is kept moist enough for the roots to actually absorb it, will greatly help the plant mount a stronger, natural defense against the deadly fungal toxins.
- **Irrigation Placement:** When installing modern drip irrigation systems, make absolutely sure the water emitters are placed far away from the actual wooden trunk. Dripping water directly onto the base of the trunk keeps the bark constantly wet, creating a perfect, swampy, humid environment for wood-rotting fungi to thrive and multiply. Keep the root crown completely dry and force the underground feeder roots to reach out into the soil for the moisture.
- **Establish Strong Roots First:** When planting brand new baby vines, do not be greedy for an early harvest. Purposefully delay allowing the vines to produce any fruit for the first few years. Forcing a tiny, young vine to ripen heavy bunches of grapes steals all its vital energy away from growing a deep, robust root system. A weak, shallow root system equals severe water stress, and early stress equals young vine decline (Petri disease). Cut the early spring flowers off with your shears and let the plant establish its deep underground foundation first. Furthermore, completely avoid using restrictive plastic "grow tubes" on young trunks. While they protect from animals, they can easily create a humid, wet microclimate that encourages rapid, weak shoot growth over strong, essential root development.

- **Proper Diagnosis:** Finally, do not confuse Esca with other common vineyard problems before you start cutting down your plants. For example, if you see holes and black marks on the leaves and green shoots, it might just be a surface disease like Anthracnose, or even physical damage from a severe summer hail storm. Hail damage is usually only visible on the top side of the shoot (where the falling ice physically hit the plant), whereas spreading fungal diseases are scattered everywhere. Always make sure you take a saw and carefully cut into the wood to look for the undeniable dark internal spots and rings to truly confirm you are battling Esca before you start undertaking drastic trunk surgery.

7. References / Sources

- Michigan State University Extension - Grapevine decline or Esca
- UC IPM - Esca (Black Measles) Symptoms
- Pacific Northwest Pest Management Handbooks - Grape (Vitis spp.)-Esca, Young Esca, and Petri Disease
- ORCAL Inc - Grape Diseases & Pests
- Phytopathologia Mediterranea - Esca (black measles) of grapevine. An overview
- PMC - Biostimulant AF5 Field Trials
- Vineyard Team - Grapevine Trunk Diseases Prevention and Management
- NC State University - Grapevine Trunk Disease Management
- Pacific Northwest Pest Management Handbooks - Symptoms * International Wine Challenge - How to deal with tinder/Esca
- PMC - Esca of grapevine complex
- Advanced Viticulture - Esca overview and tiger striping
- Pacific Northwest Pest Management Handbooks - Cultural Control Practices
- UC IPM - Esca Management and Overwintering
- PMC - AF5 biostimulant physiological basis
- ResearchGate - Susceptibility of grape varieties to esca disease
- UC IPM - Esca foliar and fruit symptoms
- PMC - Genetic Loci for disease resistance
- BIO Web of Conferences - Varietal susceptibility to Esca
- VitiSeal - Natural organic pruning wound sealants
- Vineyard Team - Pest Management / Eutypa & Esca
- Wax and Grafts - InocBloc organic pine tar
- Vine Works - Dealing with pruning wounds
- Koanga Institute - Tree paste recipe
- Vidacycle - Robin Snowdon's anti-microbial pruning paste
- YouTube - Making an organic pruning paste (beeswax/linseed)
- Kat's Garden - How to make your own pruning paste
- Friendly Haven - Biodynamic tree paste
- Pacific Northwest Pest Management Handbooks - Irrigation Management
- Vineyard Team - The Importance of Post-Harvest Irrigation * MDPI - Esca disease and fertilizer applications
- Advanced Viticulture - Tyloses and environmental stressors
- CDMS - Rally fungicide instructions
- UGA Viticulture - Topsin use for dormant pruning wounds
- FMC - Grapevine Trunk Disease Tech Sheet (Spore dispersal)

- Michigan State University Extension - Preventing grapevine decline (Lime sulfur)
- UPL Corp - Pointers for pruning wound protection
- Wine-Blog - Sustainable Viticulture (VitiSeal)
- VitiSeal - Product overview
- UPL New Zealand - VitiSeal application rates * Pacific Northwest Pest Management Handbooks - Symptom severity and water stress
- ResearchGate - Nitrogen nutrition impacts grapevine esca leaf symptom incidence
- MDPI - Macro and microelements effect on Esca
- BioRxiv - Stomatal conductance and Esca
- UConn Soil Testing - Suggested fertilizer practices for grapes
- PMC - AF5 mitigation strategies
- Pacific Northwest Pest Management Handbooks - Comprehensive Esca Guide

8. Key Takeaways

- **Esca is a Complex Wood Decay Syndrome:** It is absolutely not a single, simple disease. It is a slow succession of multiple, different fungal species that invade the trunk, heavily clog the plant's water pipes with dark gummy spots, and eventually rot the hard structural wood into a soft, spongy white mass over many years.
- **Recognize the Warning Signs:** Look carefully for the classic, brightly colored "tiger stripe" pattern on the leaves (yellow or red borders with dead brown centers) and "black measles" (dark purple spots) on the grapes themselves. In extremely hot summers following rain, a mature vine may suffer "apoplexy" and suddenly collapse and die within a matter of days.
- **Spores Enter Through Open Pruning Cuts:** The disease primarily spreads during the cold, wet winter months when rain and wind blow microscopic fungal spores directly into fresh, open, bleeding cuts made by vineyard pruning shears.
- **Plant Stress is the Ultimate Trigger:** The fungi can live silently inside the vine as endophytes for many years without causing any harm. Deadly symptoms usually only explode when the vine is severely stressed by drought, poor soil compaction, heavy fruit crop loads, or sudden summer heatwaves.
- **Protect Your Wounds Immediately:** The single most important management practice is sealing fresh pruning wounds within 24 hours of cutting. Growers can use homemade natural pastes (clay, cow manure, beeswax), commercial organic sealants (VitiSeal, pine tar), living biocontrols (Trichoderma fungi), or systemic chemical fungicides (Topsin M and Rally).
- **Change Your Pruning Habits:** Practice delayed pruning (waiting until warmer spring weather so cuts heal in a few days instead of several weeks) or double pruning (cutting high in the winter and making the final clean cut in the spring) to drastically reduce the window of infection risks.
- **Trunk Renewal Can Save the Vine:** If a vine is badly infected, you do not have to completely rip out the root system. You can saw off the diseased trunk near the ground until you reach clean white wood, and train a highly vigorous, healthy new green shoot to take its place.
- **Avoid Excessive Nitrogen Fertilizers:** Over-fertilizing a sick vine with nitrogen creates a massive leaf canopy that demands far too much water. Because the trunk is clogged, this causes the vine to stress out and triggers severe Esca leaf symptoms. Use balanced

10-10-10 fertilizers and ensure the vine is thoroughly well-watered after the autumn harvest to keep it strong for the winter.

- **Choose the Right Grape Variety:** If planting a new vineyard in high-risk, humid areas, purposefully select grape varieties with naturally strong wood and incredibly narrow water vessels (like Merlot) rather than highly susceptible types with wide vessels (like Cabernet Sauvignon). Keep all tools meticulously sanitized and only buy certified disease-free baby plants from highly reputable nurseries.

Works cited

1. Grapevine decline or Esca - Integrated Pest Management, https://www.canr.msu.edu/ipm/diseases/grapevine_decline_or_esca
2. Factors Involved on Tiger-Stripe Foliar Symptom Expression of Esca of Grapevine - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC8224345/>
3. Grape (Vitis spp.)-Esca, Young Esca, and Petri Disease, <https://pnwhandbooks.org/plantdisease/host-disease/grape-vitis-spp-esca-young-esca-petri-disease>
4. View of Esca (Black Measles) of Grapevine. An Overview | Phytopathologia Mediterranea - FUPRESS, <https://oajournals.fupress.net/index.php/pm/article/view/4785/4783>
5. Esca (Black Measles) / Grape / Agriculture: Pest Management ..., <https://ipm.ucanr.edu/agriculture/grape/esca-black-measles/>
6. "Eutypa" A.K.A. Grapevine Trunk Disease - Vineyard Team, <https://www.vineyardteam.org/resources/resource-library/pest-management.php?id=664>
7. Esca and Young Vine Decline: what you should know - Advanced Viticulture, <https://advancedvit.com/esca/>
8. Grape Diseases & Pests | ORCAL Crop Protection, <https://orcalinc.com/grape-diseases-pests>
9. How to deal with tinder - International Wine Challenge | IWC, <https://www.internationalwinechallenge.com/Canopy-Articles/how-to-deal-with-tinder.html>
10. GRAPEVINE TRUNK DISEASE:, https://ag.fmc.com/us/sites/default/files/2023-10/23-FMC-0163%20Grapevine%20Trunk%20Disease%20Tech%20Sheet%20Update_053023_VF.pdf
11. DEALING WITH PRUNING WOUNDS - VineWorks, <https://vine-works.com/blogs/advice/dealing-with-pruning-wounds>
12. Grapevine Trunk Diseases, Current Management Options - Vineyard Team, [https://www.vineyardteam.org/files/resources/Grapevine%20Trunk%20Diseases_Final%2010.14.2022%20\(1\).pdf](https://www.vineyardteam.org/files/resources/Grapevine%20Trunk%20Diseases_Final%2010.14.2022%20(1).pdf)
13. Biodynamic Tree Paste - Friendly Haven Rise Farm, <https://friendlyhaven.com/biodynamic-tree-paste/>
14. Tree Paste Recipe | Kōanga, <https://koanga.org.nz/knowledge-base/tree-paste-recipe/>
15. Recipe: Robin Snowdon's anti-microbial pruning paste - Sectormentor - Vidacycle, <https://vines.vidacycle.com/blog/recipe-robin-snowdons-anti-microbial-pruning-paste/>
16. How To Make Organic Pruning Paste & Organic Hand Cream - YouTube, <https://www.youtube.com/watch?v=myamYQntJkc>
17. How to make your own pruning paste - Kat's Garden, <https://kats-garden.nz/blog/pruning-paste>
18. vitiseal: HOME, <https://www.vitiseal.com/>
19. Vitiseal Label, https://nz.uplcorp.com/download_links/Vitiseal_Label_March2023.pdf
20. Sustainable Viticulture ~ Vine Health Made Simple - Wine Blog, <https://www.wine-blog.org/index.php/2013/01/30/sustainable-viticulture-vine-health-made-simple>
21. InocBloc® Organic Wound Protectant : a must-have for vine and fruit tree growing - Wax and Grafts, <http://www.waxandgrafts.com/news/2022/2/16/inocbloc-organic-pine-tar-a-must-have-for-vine-a>

nd-fruit-tree-growing 22. Trunk Diseases in Grapes,
https://grapes.ces.ncsu.edu/documents/642376/Hoffmann_Surry_TrunkDisease_2019.pdf 23. Mitigating grapevine esca disease: an innovative integrated management strategy to reduce incidence and severity by enhancing plant physiology and defence mechanisms - PMC,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12934101/> 24. A Few Pointers for Pruning Wound Protection - UPL Corp,
<https://www.uplcorp.com/us/blog-details/pointers-pruning-wound-protection> 25. Recommendation, <https://www.cdms.telusagcg.com/ldat/ld4EK010.pdf> 26. Preventing grapevine decline in Michigan: Pruning wound management and best vineyard practices - Grapes - MSU College of Agriculture and Natural Resources,
<https://www.canr.msu.edu/news/preventing-grapevine-decline-in-michigan> 27. Topsin Use for Dormant Pruning Wounds - Viticulture - UGA,
<https://viticulture.uga.edu/2021/01/topsin-use-for-dormant-pruning-wounds/> 28. Susceptibility of grape varieties to esca disease - BIO Web of Conferences,
https://www.bio-conferences.org/articles/bioconf/pdf/2016/02/bioconf-oiv2016_01041.pdf 29. Susceptibility of grape varieties to esca disease - ResearchGate,
https://www.researchgate.net/publication/309475023_Susceptibility_of_grape_varieties_to_esca_disease 30. Information on disease resistance patterns of grape varieties may improve disease management - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC9704053/> 31. The Importance of Post-Harvest Irrigation and Fertilization - Vineyard Team,
<https://www.vineyardteam.org/files/resources/Irrigation%20&%20Fertilization.pdf> 32. Effect of Different Foliar Fertilizer Applications on Esca Disease of Grapevine: Symptom Expression and Nutrient Content in the Leaf and Composition of the Berry - MDPI,
<https://www.mdpi.com/2073-4395/13/5/1355> 33. (PDF) Nitrogen nutrition impacts grapevine esca leaf symptom incidence, physiology and metabolism. - ResearchGate,
https://www.researchgate.net/publication/383683133_Nitrogen_nutrition_impacts_grapevine_esca_leaf_symptom_incidence_physiology_and_metabolism 34. Nitrogen nutrition impacts grapevine esca leaf symptom incidence, physiology and metabolism - bioRxiv,
<https://www.biorxiv.org/content/biorxiv/early/2024/09/02/2024.08.31.610625.full.pdf> 35. Suggested Fertilizer Practices For Grapes | Soil Nutrient Analysis Laboratory | College of Agriculture, Health and Natural Resources | University of Connecticut,
<https://soiltesting.cahnrc.uconn.edu/suggested-fertilizer-practices-for-grapes/>